

Sean Graham Week 2

1.

The screenshot shows the RStudio interface. The top pane displays a data frame named 'SP500' with 1,773 rows and 6 columns. The columns are labeled: Date, SP500, Dividend, Earnings, CPI, and Rate. The bottom pane shows the R console with the following output:

```
[workspace loaded from ~/.RData]
> library(readxl)
> SP500 <- read_excel("SP500 (1).xls", sheet = "Data")
> view(SP500)
> |
```

2. There are 1,773 rows and 6 columns

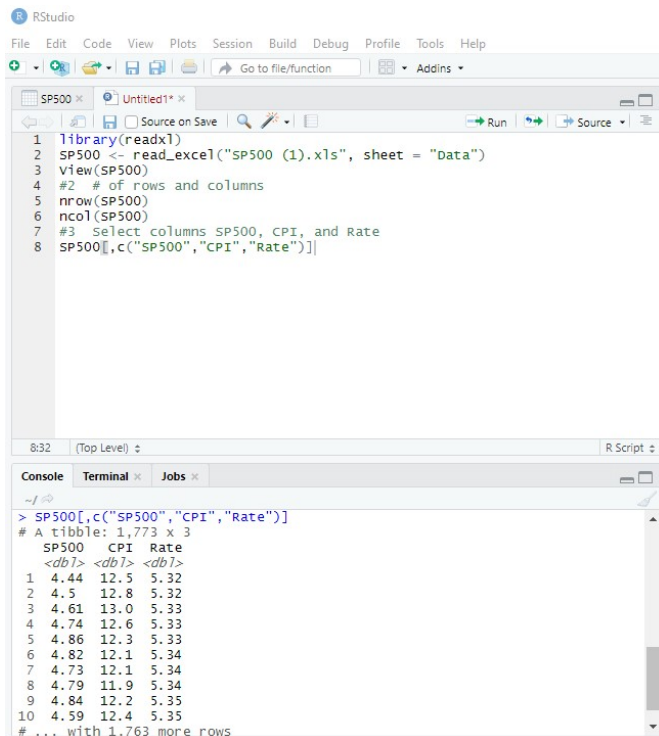
The screenshot shows the RStudio interface with the R script editor open. The script contains the following code:

```
1 library(readxl)
2 SP500 <- read_excel("SP500 (1).xls", sheet = "Data")
3 view(SP500)
4 #2 # of rows and columns
5 nrow(SP500)
6 ncol(SP500)
```

The bottom pane shows the R console with the following output:

```
[workspace loaded from ~/.RData]
> library(readxl)
> SP500 <- read_excel("SP500 (1).xls", sheet = "Data")
> view(SP500)
> nrow(SP500)
[1] 1773
> ncol(SP500)
[1] 6
> |
```

3.



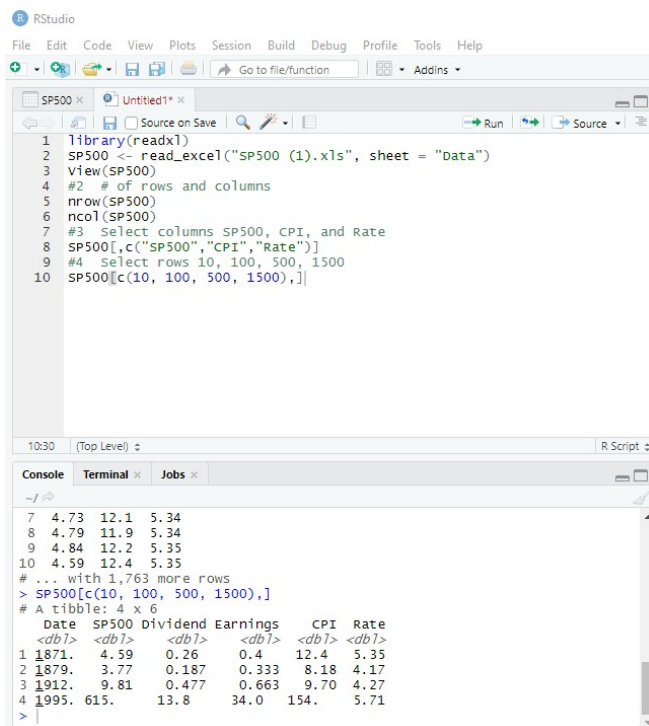
RStudio interface showing the following code in the editor:

```
1 library(readxl)
2 SP500 <- read_excel("SP500 (1).xls", sheet = "Data")
3 View(SP500)
4 #2 # of rows and columns
5 nrow(SP500)
6 ncol(SP500)
7 #3 Select columns SP500, CPI, and Rate
8 SP500[,c("SP500","CPI","Rate")]
```

The console output shows the result of the command:

```
> SP500[,c("SP500","CPI","Rate")]
# A tibble: 1,773 x 3
  SP500    CPI Rate
<dbl> <dbl> <dbl>
1  4.44  12.5  5.32
2  4.5   12.8  5.32
3  4.61  13.0  5.33
4  4.74  12.6  5.33
5  4.86  12.3  5.33
6  4.82  12.1  5.34
7  4.73  12.1  5.34
8  4.79  11.9  5.34
9  4.84  12.2  5.35
10 4.59  12.4  5.35
# ... with 1,763 more rows
```

4.



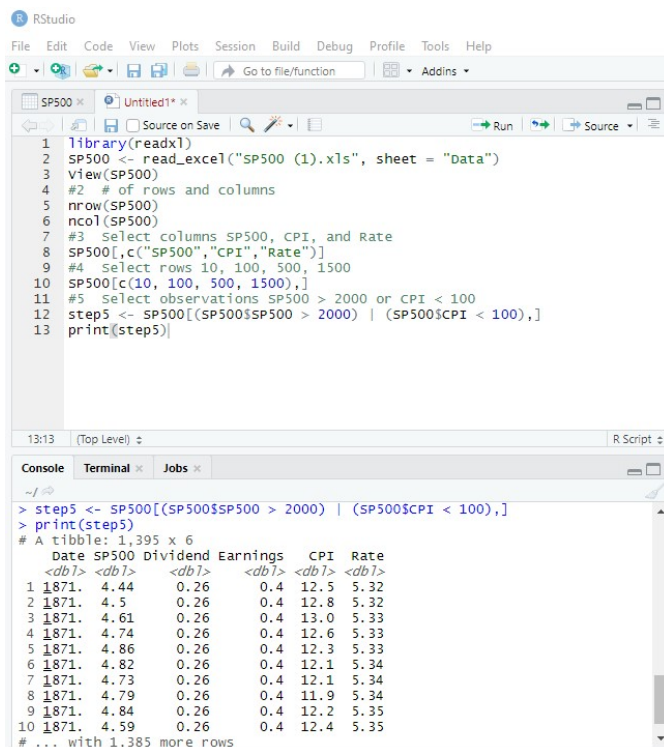
RStudio interface showing the following code in the editor:

```
1 library(readxl)
2 SP500 <- read_excel("SP500 (1).xls", sheet = "Data")
3 View(SP500)
4 #2 # of rows and columns
5 nrow(SP500)
6 ncol(SP500)
7 #3 Select columns SP500, CPI, and Rate
8 SP500[,c("SP500","CPI","Rate")]
9 #4 Select rows 10, 100, 500, 1500
10 SP500[c(10, 100, 500, 1500),]
```

The console output shows the result of the command:

```
> SP500[c(10, 100, 500, 1500),]
# A tibble: 4 x 6
  Date      SP500 Dividend Earnings    CPI Rate
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1871.    4.59    0.26    0.4  12.4  5.35
2 1879.    3.77    0.187  0.333  8.18  4.17
3 1912.    9.81    0.477  0.663  9.70  4.27
4 1995.   615.   13.8   34.0  154.  5.71
```

5.



RStudio interface showing the execution of R code. The script editor contains 13 lines of code. The console shows the output of the code, including a tibble with 1,395 rows and 6 columns: Date, SP500, Dividend, Earnings, CPI, and Rate. The first 10 rows are displayed.

```

1 library(readxl)
2 SP500 <- read_excel("SP500 (1).xls", sheet = "Data")
3 view(SP500)
4 #2 # of rows and columns
5 nrow(SP500)
6 ncol(SP500)
7 #3 Select columns SP500, CPI, and Rate
8 SP500[,c("SP500","CPI","Rate")]
9 #4 Select rows 10, 100, 500, 1500
10 SP500[c(10, 100, 500, 1500),]
11 #5 Select observations SP500 > 2000 or CPI < 100
12 step5 <- SP500[(SP500$SP500 > 2000) | (SP500$CPI < 100),]
13 print(step5)

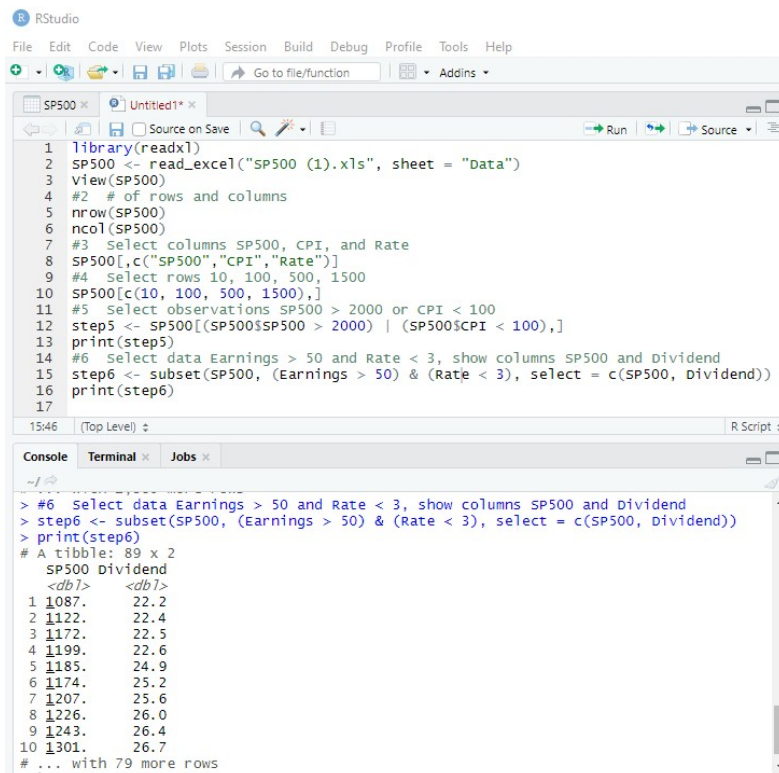
```

```

~/ > step5 <- SP500[(SP500$SP500 > 2000) | (SP500$CPI < 100),]
> print(step5)
# A tibble: 1,395 x 6
  Date      SP500 Dividend Earnings  CPI Rate
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1871.    4.44    0.26    0.4  12.5  5.32
2 1871.    4.5    0.26    0.4  12.8  5.32
3 1871.    4.61   0.26    0.4  13.0  5.33
4 1871.    4.74   0.26    0.4  12.6  5.33
5 1871.    4.86   0.26    0.4  12.3  5.33
6 1871.    4.82   0.26    0.4  12.1  5.34
7 1871.    4.73   0.26    0.4  12.1  5.34
8 1871.    4.79   0.26    0.4  11.9  5.34
9 1871.    4.84   0.26    0.4  12.2  5.35
10 1871.    4.59   0.26    0.4  12.4  5.35
# ... with 1,385 more rows

```

6.



RStudio interface showing the execution of R code. The script editor contains 17 lines of code. The console shows the output of the code, including a tibble with 89 rows and 2 columns: SP500 and Dividend. The first 10 rows are displayed.

```

1 library(readxl)
2 SP500 <- read_excel("SP500 (1).xls", sheet = "Data")
3 view(SP500)
4 #2 # of rows and columns
5 nrow(SP500)
6 ncol(SP500)
7 #3 Select columns SP500, CPI, and Rate
8 SP500[,c("SP500","CPI","Rate")]
9 #4 Select rows 10, 100, 500, 1500
10 SP500[c(10, 100, 500, 1500),]
11 #5 Select observations SP500 > 2000 or CPI < 100
12 step5 <- SP500[(SP500$SP500 > 2000) | (SP500$CPI < 100),]
13 print(step5)
14 #6 Select data Earnings > 50 and Rate < 3, show columns SP500 and Dividend
15 step6 <- subset(SP500, (Earnings > 50) & (Rate < 3), select = c(SP500, Dividend))
16 print(step6)
17

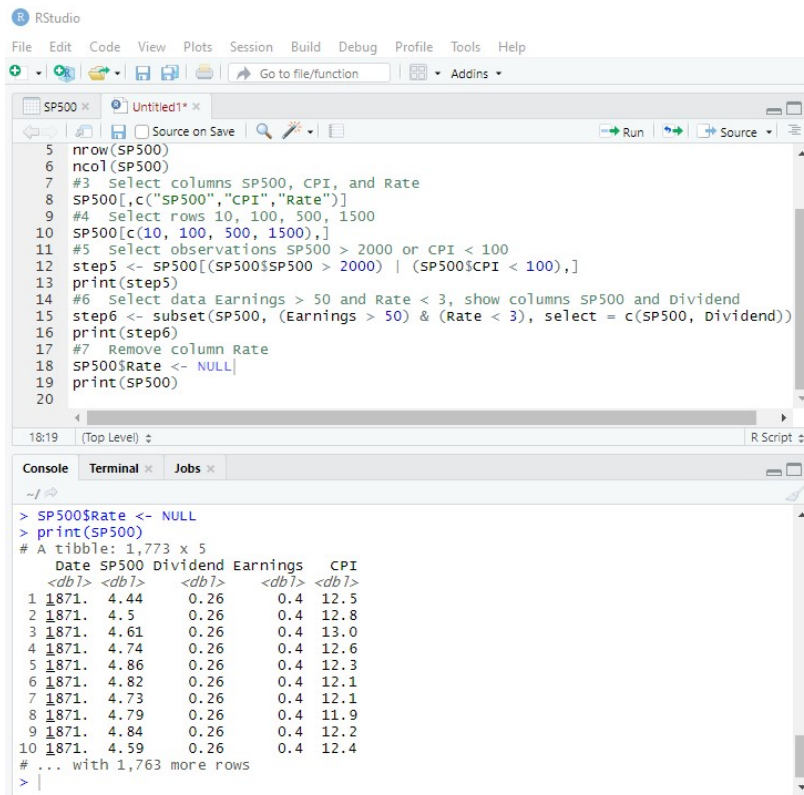
```

```

~/ > #6 Select data Earnings > 50 and Rate < 3, show columns SP500 and Dividend
> step6 <- subset(SP500, (Earnings > 50) & (Rate < 3), select = c(SP500, Dividend))
> print(step6)
# A tibble: 89 x 2
  SP500 Dividend
<dbl> <dbl>
1 1087.    22.2
2 1122.    22.4
3 1172.    22.5
4 1199.    22.6
5 1185.    24.9
6 1174.    25.2
7 1207.    25.6
8 1226.    26.0
9 1243.    26.4
10 1301.    26.7
# ... with 79 more rows

```

7.



RStudio interface showing R code for data manipulation. The code defines a data frame 'SP500' with columns 'Date', 'SP500', 'Dividend', 'Earnings', and 'CPI'. It filters for observations where SP500 > 2000 or CPI < 100, then selects columns 'SP500' and 'Dividend' for observations where Earnings > 50 and Rate < 3. The console output shows the resulting tibble with 1,773 rows and 5 columns.

```

5 nrow(SP500)
6 ncol(SP500)
7 #3 Select columns SP500, CPI, and Rate
8 SP500[,c("SP500", "CPI", "Rate")]
9 #4 Select rows 10, 100, 500, 1500
10 SP500[c(10, 100, 500, 1500),]
11 #5 Select observations SP500 > 2000 or CPI < 100
12 step5 <- SP500[(SP500$SP500 > 2000) | (SP500$CPI < 100),]
13 print(step5)
14 #6 Select data Earnings > 50 and Rate < 3, show columns SP500 and Dividend
15 step6 <- subset(SP500, (Earnings > 50) & (Rate < 3), select = c(SP500, Dividend))
16 print(step6)
17 #7 Remove column Rate
18 SP500$Rate <- NULL
19 print(SP500)
20

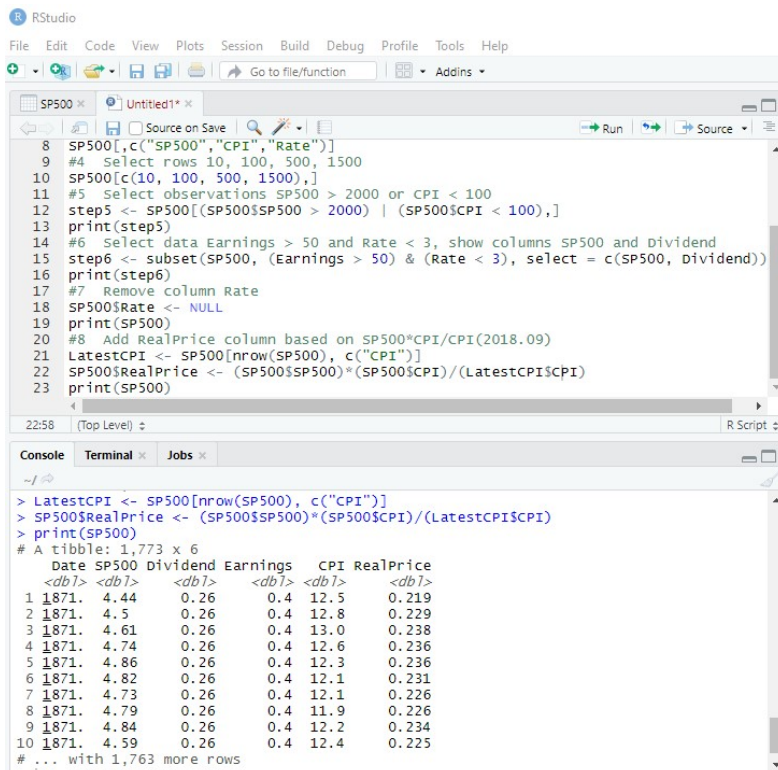
```

```

> SP500$Rate <- NULL
> print(SP500)
# A tibble: 1,773 x 5
  Date      SP500 Dividend Earnings   CPI
<dbl> <dbl> <dbl> <dbl> <dbl>
1 1871.  4.44    0.26    0.4  12.5
2 1871.  4.5     0.26    0.4  12.8
3 1871.  4.61    0.26    0.4  13.0
4 1871.  4.74    0.26    0.4  12.6
5 1871.  4.86    0.26    0.4  12.3
6 1871.  4.82    0.26    0.4  12.1
7 1871.  4.73    0.26    0.4  12.1
8 1871.  4.79    0.26    0.4  11.9
9 1871.  4.84    0.26    0.4  12.2
10 1871.  4.59    0.26    0.4  12.4
# ... with 1,763 more rows
>

```

8.



RStudio interface showing R code for data manipulation. The code defines a data frame 'SP500' with columns 'Date', 'SP500', 'Dividend', 'Earnings', and 'CPI'. It filters for observations where SP500 > 2000 or CPI < 100, then selects columns 'SP500' and 'Dividend' for observations where Earnings > 50 and Rate < 3. The console output shows the resulting tibble with 1,773 rows and 6 columns, including a new 'RealPrice' column calculated as (SP500\$SP500) * (SP500\$CPI) / (LatestCPI\$CPI).

```

8 SP500[,c("SP500", "CPI", "Rate")]
9 #4 Select rows 10, 100, 500, 1500
10 SP500[c(10, 100, 500, 1500),]
11 #5 Select observations SP500 > 2000 or CPI < 100
12 step5 <- SP500[(SP500$SP500 > 2000) | (SP500$CPI < 100),]
13 print(step5)
14 #6 Select data Earnings > 50 and Rate < 3, show columns SP500 and Dividend
15 step6 <- subset(SP500, (Earnings > 50) & (Rate < 3), select = c(SP500, Dividend))
16 print(step6)
17 #7 Remove column Rate
18 SP500$Rate <- NULL
19 print(SP500)
20 #8 Add RealPrice column based on SP500*CPI/CPI(2018.09)
21 LatestCPI <- SP500[nrow(SP500), c("CPI")]
22 SP500$RealPrice <- (SP500$SP500)*(SP500$CPI)/(LatestCPI$CPI)
23 print(SP500)

```

```

> LatestCPI <- SP500[nrow(SP500), c("CPI")]
> SP500$RealPrice <- (SP500$SP500)*(SP500$CPI)/(LatestCPI$CPI)
> print(SP500)
# A tibble: 1,773 x 6
  Date      SP500 Dividend Earnings   CPI RealPrice
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1871.  4.44    0.26    0.4  12.5    0.219
2 1871.  4.5     0.26    0.4  12.8    0.229
3 1871.  4.61    0.26    0.4  13.0    0.238
4 1871.  4.74    0.26    0.4  12.6    0.236
5 1871.  4.86    0.26    0.4  12.3    0.236
6 1871.  4.82    0.26    0.4  12.1    0.231
7 1871.  4.73    0.26    0.4  12.1    0.226
8 1871.  4.79    0.26    0.4  11.9    0.226
9 1871.  4.84    0.26    0.4  12.2    0.234
10 1871.  4.59    0.26    0.4  12.4    0.225
# ... with 1,763 more rows
>

```

9.

```

# SP500
# ... with 1,763 more rows
> #9 Add RealEarnings column based on Earnings*CPI/CPI(2018.09)
> SP500$RealEarnings <- (SP500$Earnings)*(SP500$CPI)/(LatestCPI$CPI)
> print(SP500)
# A tibble: 1,773 x 7
  Date SP500 Dividend Earnings CPI RealPrice RealEarnings
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1871. 4.44 0.26 0.4 12.5 0.219 0.0192
2 1871. 4.5 0.26 0.4 12.8 0.229 0.0204
3 1871. 4.61 0.26 0.4 13.0 0.238 0.0207
4 1871. 4.74 0.26 0.4 12.6 0.236 0.0199
5 1871. 4.86 0.26 0.4 12.3 0.236 0.0194
6 1871. 4.82 0.26 0.4 12.1 0.231 0.0191
7 1871. 4.73 0.26 0.4 12.1 0.226 0.0191
8 1871. 4.79 0.26 0.4 11.9 0.226 0.0188
9 1871. 4.84 0.26 0.4 12.2 0.234 0.0193
10 1871. 4.59 0.26 0.4 12.4 0.225 0.0196
# ... with 1,763 more rows

```

10.

```

# SP500
# ... with 1,763 more rows
> #10 Add PERatio column based on RealPrice/RealEarnings
> SP500$PERatio <- (SP500$RealPrice)/(SP500$RealEarnings)
> print(SP500)
# A tibble: 1,773 x 8
  Date SP500 Dividend Earnings CPI RealPrice RealEarnings PERatio
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1871. 4.44 0.26 0.4 12.5 0.219 0.0192 11.1
2 1871. 4.5 0.26 0.4 12.8 0.229 0.0204 11.2
3 1871. 4.61 0.26 0.4 13.0 0.238 0.0207 11.5
4 1871. 4.74 0.26 0.4 12.6 0.236 0.0199 11.9
5 1871. 4.86 0.26 0.4 12.3 0.236 0.0194 12.1
6 1871. 4.82 0.26 0.4 12.1 0.231 0.0191 12.0
7 1871. 4.73 0.26 0.4 12.1 0.226 0.0191 11.8
8 1871. 4.79 0.26 0.4 11.9 0.226 0.0188 12.0
9 1871. 4.84 0.26 0.4 12.2 0.234 0.0193 12.1
10 1871. 4.59 0.26 0.4 12.4 0.225 0.0196 11.5
# ... with 1,763 more rows

```